

CS701 Software Architecture & Design Patterns Notes

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Unit 1: Architectural Foundations & SOLID Principles

SOLID Design Principles:

- Single Responsibility: A class should have only one reason to change.
- Open/Closed: Software entities should be open for extension but closed for modification.
- Liskov Substitution: Subtypes must be substitutable for their base types.
- Interface Segregation: Clients should not be forced to depend upon interfaces they do not use.
- Dependency Inversion: Depend upon abstractions, not concretions.

Unit 2: Core Architectural Patterns

1. Layered (N-Tier) Architecture: Separates presentation, business logic, persistence, and database layers.
2. Event-Driven Architecture: Decoupled publishers and subscribers reacting asynchronously to domain events via message brokers (Kafka, RabbitMQ).
3. Microservices Architecture: Decentralized, independently deployable services communicating via REST APIs or gRPC, each managing its own isolated datastore.

